

Random dynamical systems

A discrete-time random map

$$F_W^t := F_{W_t} \circ \dots \circ F_{W_1}$$

W : sequence of random variables
chosen iid $\{W_1, \dots, W_t, \dots\}$

Continuous-time :

$$dX_t = f(X_t)dt + \Sigma(X_t)dW_t$$

Usually considered in Stratonovich form
Then for almost every path W ,

flow map $\varphi^{t+u}(x; W)$

$$= \varphi_t(\varphi_u(x; W); \sigma_u(W))$$

σ_u : shift of ^{path} sequence W .
by time u .

Non-degenerate case : Noise W_t can

push the state vector in any infinitesimal direction in tangent space.

More generally,

$$dX_t = f(X_t)dt + \sum_{i=1}^m \sum_t^i dW_t^i$$

dW_t^i : scalar Brownian motion

$\sum_t^i(X_t)$: vector field in $\mathbb{R}^d$

m : no of Brownian motions

$m < n$: degenerate noise

W_t^i : standard Brownian motion

$$\bullet \lim_{n \rightarrow \infty} \frac{S_n}{\sqrt{n}} \xrightarrow{d} W_t^i$$

where $S_n = \sum_{i=1}^n B_i$ $B_i = \pm 1$ with prob. $\frac{1}{2}$.

That is the sum of coin tosses with scaling converges in distribution to a random variable called Brownian motion

$$W_t \sim N(0, t)$$

$W_t - W_s \sim N(0, t-s)$ independent increments.

Ito form of Stratonovich process:

$$dX_t = f(X_t) dt + \Sigma(X_t) dW_t$$

$$+ \frac{1}{2} \sum_j \sum_i \nabla_j \Sigma_i(X_t) dW_t$$